

Independent Study Plan

Building Language Models Using Reinforcement Learning

A 12-Week Research Project with Publication Goals

Executive Summary

This independent study explores a novel approach to training Large Language Models using Reinforcement Learning (RL) techniques instead of traditional supervised learning methods. The research challenges current practices by investigating whether AI systems can learn language purely through trial-and-error with appropriate feedback signals, similar to how humans might learn through interaction and reward.

Research Question: Can we build effective language models using only reinforcement learning, without requiring massive text datasets for supervised pre-training?

Potential Impact: This could lead to more sample-efficient training methods, better alignment with human preferences, and new insights into how artificial intelligence systems learn language.

Research Background and Motivation

Current State of Language Model Training

Today's large language models like GPT-4 and ChatGPT are trained in two phases:

1. **Supervised Learning Phase:** Models learn to predict the next word from billions of text examples
2. **Reinforcement Learning Phase:** Models are fine-tuned using human feedback to make them more helpful and safe

This approach has several limitations:

- **Data Requirements:** Requires enormous text datasets (trillions of words)
- **Computational Costs:** Extremely expensive to train (\$10-100 million per model)
- **Alignment Issues:** Models may learn undesirable behaviors from internet text
- **Efficiency Questions:** Unclear if massive pre-training is fundamentally necessary

Proposed Alternative Approach

This research investigates whether we can train language models using *only* reinforcement learning, where the AI system:

- Starts with no knowledge of language
- Learns through trial and error
- Receives feedback on the quality of generated text
- Gradually improves through this feedback loop

Key Technical Concepts (Explained Simply)

Reinforcement Learning (RL)

Like training a pet with treats and corrections, RL teaches AI systems through rewards and penalties. The system tries different actions, gets feedback, and learns to maximize rewards over time.

Energy-Based Models (EBMs)

Think of these as "energy landscapes" where good text has low energy (like a ball rolling to the bottom of a hill) and bad text has high energy. The model learns to generate text by finding low-energy configurations.

Hierarchical Policies

Instead of choosing one word at a time randomly, we break the problem into levels: first decide on high-level intentions (like "answer a question"), then choose specific words to fulfill that intention.

Detailed 12-Week Timeline

Weeks 1-2: Foundation Building

Week 1: Understanding Reinforcement Learning

Learning Goals: Master the basic concepts of how machines learn through trial and error

Day	Activity	Deliverable
Mon-Tue	Study RL basics: agents, environments, rewards	Summary notes on key concepts
Wed-Thu	Implement simple RL algorithm (Q-learning)	Working code for basic game (tic-tac-toe)
Fri-Sat	Advanced RL: policy gradients, actor-critic	Implementation of policy gradient method
Sun	Integration and testing	Complete RL toolkit with examples

Key Resources:

- Sutton & Barto "Reinforcement Learning: An Introduction" (Chapters 1-4)
- OpenAI Spinning Up in Deep RL (online course)
- Python libraries: Gym, Stable Baselines3

Week 1 Milestone: Successfully train an RL agent to play simple games (CartPole, FrozenLake) with >90% success rate

Week 2: Connecting RL to Language

Learning Goals: Understand how language generation can be framed as a reinforcement learning problem

Core Activities:

- Study how current language models work (transformers, attention mechanisms)
- Read key papers on using RL for text generation
- Implement basic character-level text generation using RL
- Set up computational environment and tools

Key Papers to Read:

1. "Training Language Models to Follow Instructions with Human Feedback" (OpenAI)
2. "Constitutional AI: Harmlessness from AI Feedback" (Anthropic)
3. "Language Models are Few-Shot Learners" (GPT-3 paper)

Week 2 Milestone: Working character-level text generator using RL that can learn simple patterns (like repeating "hello world")

Weeks 3-4: Research Design and Literature Review

Week 3: Comprehensive Literature Review

Objective: Understand the current state of research and identify gaps our work can fill

Research Areas to Cover:

- **RLHF (RL from Human Feedback):** How current systems use RL to improve language models
- **Pure RL for Text:** Previous attempts to generate text using only RL
- **Energy-Based Models:** Mathematical frameworks for modeling probability distributions
- **Language Model Architecture:** How transformers and attention mechanisms work

Deliverable: 15-page literature review identifying:

- What has been tried before and why it didn't work well
- Current limitations of existing approaches
- Opportunities for our novel approach
- Theoretical foundations for our method

Week 4: Mathematical Framework Development

Objective: Develop rigorous mathematical foundations for our approach

Core Components to Formalize:

1. **Problem Definition:** Precisely define language generation as an RL problem
2. **Hierarchical Approach:** Mathematical framework for multi-level decision making
3. **Energy-Based Values:** Connect energy models to value functions in RL

4. **Reward Design:** Formal specification of what constitutes "good" text

Weeks 3-4 Milestone: Complete research proposal with mathematical specifications, literature review, and theoretical analysis proving the approach is theoretically sound

Weeks 5-6: Core Implementation Phase

Week 5: Environment and Reward System

Objective: Build the "training ground" where our AI will learn language

Implementation Tasks:

1. **Text Generation Environment:** A system where the AI can generate text and receive feedback
2. **Multi-Component Rewards:** Feedback system that evaluates text quality on multiple dimensions:
 - Fluency: How natural does the text sound?
 - Coherence: Does the text make logical sense?
 - Task Completion: Does it accomplish the intended goal?
 - Safety: Is the content appropriate and non-harmful?
3. **Task-Specific Environments:** Specialized training setups for different language tasks

Week 6: Hierarchical Policy Implementation

Objective: Build an AI system that thinks at multiple levels when generating text

Key Innovation: Instead of choosing words randomly, the system first decides on high-level intentions (like "I want to answer this question about history"), then chooses specific words to fulfill that intention.

Technical Components:

- **High-Level Policy:** Decides on semantic intentions and goals
- **Low-Level Policy:** Chooses specific words based on current intentions
- **Training Algorithm:** Method to train both levels simultaneously

Weeks 5-6 Milestone: Working hierarchical system that can generate coherent text for simple tasks like question-answering and basic conversation

Weeks 7-8: Advanced Technical Implementation

Week 7: Energy-Based Value Functions

Objective: Implement the core theoretical innovation connecting energy models to reinforcement learning

Key Insight: The same neural network can both evaluate how good a piece of text is AND generate new text by finding low-energy (high-quality) configurations.

Technical Implementation:

- **Energy Network:** Neural network that assigns "energy scores" to text
- **Value Function Integration:** Use energy scores as value estimates for RL
- **Text Generation:** Generate text by finding low-energy configurations

- **Training Loop:** Simultaneously improve both evaluation and generation

Week 8: Multi-Scale Learning Integration

Objective: Enable the system to learn language structure at multiple time scales simultaneously

Components:

- **Short-term Learning:** Immediate word-to-word dependencies
- **Medium-term Learning:** Sentence-level coherence
- **Long-term Learning:** Document-level structure and narrative flow
- **Structural Learning:** Grammar and syntax discovery without explicit teaching

Weeks 7-8 Milestone: Integrated system showing clear improvements in text quality, with evidence of learning linguistic structure without explicit supervision

Weeks 9-10: Experimental Validation

Week 9: Benchmark Testing

Objective: Rigorously test our approach against established benchmarks and baselines

Task Category	Specific Test	Success Metric
Question Answering	SQuAD dataset	F1 score $\geq 70\%$
Text Summarization	CNN/DailyMail	ROUGE score competitive with baselines
Code Generation	Simple programming tasks	Functional code generation
Dialogue	Conversational tasks	Human evaluation scores

Comparison Baselines:

- Standard supervised learning approaches (GPT-2, T5)
- Current RLHF methods (InstructGPT-style models)
- Hybrid approaches (pre-training + RL)

Week 10: Deep Analysis and Interpretation

Objective: Understand how and why our approach works through detailed analysis

Analysis Components:

- **Representation Learning:** What linguistic knowledge has the system discovered?
- **Sample Efficiency:** How much training data do we need compared to traditional methods?
- **Energy Landscape Visualization:** How does the system organize different types of text?

- **Failure Analysis:** What are the limitations and failure modes?

Weeks 9-10 Milestone: Comprehensive experimental results showing competitive performance on at least 2 out of 4 benchmark tasks, with clear evidence of the approach's advantages and limitations

Weeks 11-12: Research Paper and Publication

Week 11: Large-Scale Experiments and Paper Writing

Objective: Scale up experiments and begin writing the research paper

Final Experimental Push:

- **Scaling Experiments:** Train larger models to demonstrate scalability
- **Additional Baselines:** Compare with state-of-the-art systems
- **Statistical Validation:** Ensure all results are statistically significant
- **Ablation Studies:** Prove that each component of our approach contributes to success

Paper Structure:

1. **Introduction:** Problem motivation and our contributions
2. **Related Work:** How our work fits into existing research
3. **Methodology:** Detailed explanation of our approach
4. **Experiments:** Comprehensive evaluation and results
5. **Analysis:** Deep dive into why the approach works
6. **Conclusion:** Implications and future work

Week 12: Paper Finalization and Submission

Objective: Complete and submit a publication-quality research paper

Target Venues:

Conference	Deadline	Focus	Acceptance Rate
NeurIPS 2025	May 2025	Machine Learning Theory & Practice	~25%
ICML 2025	Jan 2025	Machine Learning Methods	~28%
EMNLP 2025	June 2025	Natural Language Processing	~25%

Week 12 Milestone: Complete, submission-ready research paper with all supporting materials (code, data, supplementary materials)

Learning Outcomes and Assessment

Technical Skills Development

Programming and Implementation Skills

- **Deep Learning Frameworks:** PyTorch, Hugging Face Transformers
- **Reinforcement Learning:** Stable Baselines3, custom RL implementations
- **Experimental Design:** Statistical testing, ablation studies, benchmarking
- **Large-Scale Computing:** GPU programming, distributed training

Research Methodology

- **Literature Review:** Systematic analysis of academic literature
- **Hypothesis Formation:** Developing testable research questions
- **Experimental Design:** Creating rigorous experiments with proper controls
- **Statistical Analysis:** Interpreting results and ensuring significance

Communication Skills

- **Academic Writing:** Research paper composition for top venues
- **Technical Documentation:** Code documentation and reproducibility
- **Visual Communication:** Creating figures and plots to explain results
- **Presentation Skills:** Defending research findings

Assessment Methods

Assessment Component	Weight	Description	Due Date
Literature Review	15%	Comprehensive survey of related work	Week 3
Technical Implementation	25%	Working code with documentation	Week 8
Experimental Results	20%	Rigorous evaluation and analysis	Week 10
Research Paper	25%	Publication-quality manuscript	Week 12
Presentation	15%	Final defense of research	Week 12

Weekly Progress Meetings

Format: 1-hour weekly meetings with faculty supervisor

Content:

- Progress review and obstacle discussion
- Technical problem-solving
- Research direction guidance
- Paper writing feedback

Resource Requirements

Computational Resources

Phase	Hardware Needs	Software Requirements
Weeks 1-4	Standard laptop/desktop	Python, PyTorch, Jupyter
Weeks 5-8	GPU access (RTX 3080 or better)	CUDA, distributed training libraries
Weeks 9-12	Multi-GPU cluster access	Cloud computing platforms

Library and Database Access

- IEEE Xplore and ACM Digital Library for paper access
- ArXiv for preprint papers
- Standard academic datasets (SQuAD, CNN/DailyMail, etc.)
- Large text corpora for training

Expected Challenges and Risk Mitigation

Technical Challenges

Challenge	Likelihood	Mitigation Strategy
RL training instability	High	Extensive hyperparameter tuning, curriculum learning
Poor sample efficiency	Medium	Hierarchical approaches, better reward design
Computational limitations	Medium	Cloud computing, efficient implementations
Underperformance vs baselines	Medium	Focus on specific advantages, hybrid approaches

Research Challenges

- **Novel Approach Risk:** Since this is unexplored territory, results are uncertain
- **Publication Timeline:** Academic conferences have strict deadlines
- **Reproducibility:** Ensuring others can replicate our results

Contingency Plans

- **If pure RL fails:** Investigate hybrid approaches or focus on theoretical contributions
- **If scaling issues arise:** Focus on smaller-scale validation and theoretical analysis
- **If publication timeline is tight:** Submit to later conferences or journals

Long-Term Impact and Future Directions

Potential Research Impact

- **Theoretical Contributions:** New understanding of how RL can be applied to language learning
- **Practical Applications:** More efficient training methods for language models
- **Interdisciplinary Connections:** Links between RL, NLP, and cognitive science
- **Environmental Benefits:** Potentially reduced computational requirements for training

Career Development

- **Research Skills:** Experience with cutting-edge AI research
- **Publication Record:** First-author paper at top-tier venue
- **Industry Relevance:** Experience with techniques used by leading AI companies
- **Graduate School Preparation:** Strong foundation for PhD applications

Follow-Up Research Directions

- **Multimodal Extensions:** Applying the approach to vision-language tasks
- **Scaling Studies:** Investigating behavior at very large scale
- **Theoretical Analysis:** Deeper mathematical understanding of why the approach works
- **Application Domains:** Specialized applications in code, math, or science

Success Metrics

Minimum Success: A technically sound independent study with novel insights, documented in a well-written report

Target Success: A publication-ready paper showing competitive results with clear advantages over existing methods

Exceptional Success: A paradigm-shifting contribution that influences how the field approaches language model training, with acceptance at a top-tier venue like NeurIPS